MILITARY SPECIFICATION

SOLENOIDS, ELECTRICAL (NAVAL SHIPBOARD)

1. SCOPE

1.1 Scope. This specification covers high-impact shockproof, electrical solenoids for Naval Shipboard use.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact): Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-E-917 - Electric Power Equipment, Basic Requirements For (Naval Shipboard Use).
MIL-P-15024 - Plates, Identification Information and Marking For Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-I-45208 - Inspection System Requirements.
MIL-M-81203 - Manual, Technical, In-Process Reviews, Validation, and Verification Support of.

STANDARDS

MILITARY

MIL-STD-108 - Definition of and Basic Requirements For Electric and Electronic Equipment.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-721 - Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors and Safety.
MIL-STD-761 - Electric Power, Alternating Current, For Shipboard Use, Characterization and Utilization of.
MIL-STD-785 - Requirements For Reliability Program (For System and Equipment).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION
Standard IC 1-2.42 - Industrial Control.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, New York 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Samples for first article inspection. Samples of the solenoids shall be tested as specified in 4.3 (see 6.3).

3.2 General requirements. Solenoids covered by this specification shall conform with the requirements of MIL-E-917, except as specified herein.

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3.2.1 Reliability. A reliability program in accordance with MIL-STD-785 shall be submitted to the Power Distribution Section, Naval Ship Engineering Center for approval prior to testing.

3.2.2 Maintenance (preventative). An instruction sheet on preventative maintenance, as specified in MIL-STD-721, shall be included in the technical manual (see 3.4). The instruction sheet shall contain the following minimum information:

(a) A list of any possible preventative maintenance actions.
(b) Tools, test equipment, and the time frame of each preventative maintenance action.
(c) Safety precautions to be observed during the preventative maintenance actions.

3.2.3 Shock. The solenoids shall show no signs of physical damage when subjected to the shock test for grade A, class 1, type A, lightweight equipment as specified in MIL-S-901.

3.2.4 Vibration. The solenoids shall show no signs of physical damage and shall not drop out or chatter when subjected to the type I vibration test of MIL-STD-167.

3.2.5 Enclosures. When enclosed, the solenoid enclosure shall be dripproof, watertight, submersible, or explosion proof as specified (see 6.1), in accordance with MIL-STD-108. Solenoids in explosion proof enclosures shall be tested by the Bureau of Mines.

3.2.5.1 Coils in dripproof or watertight enclosures may be encapsulated or varnish impregnated. Coils in all other enclosures (see 3.2.5) shall be encapsulated.

3.2.6 Class of insulation. The solenoids shall conform with classes A, B or H for insulation.

3.2.7 Temperature rise. The maximum temperature rise for the solenoid coils, based on 50° Centigrade (C) ambient temperature, shall be:

Class A insulation - 60°C
Class B insulation - 85°C
Class H insulation - 130°C

3.2.8 Mounting positions. The solenoids shall operate properly when mounted in the following positions:

Upright
Upside down
Inclined 90 degrees to right
Inclined 90 degrees to left

3.3 Electrical requirements.

3.3.1 Operation. Unless otherwise specified (see 6.1), the solenoid shall be capable of 1,000,000 electrical operations at rated voltage and frequency (see 3.3.2). The solenoid shall operate without hesitation on energization or deenergization. The complete operation shall take less than 1 second. When the solenoid is mounted in any of the positions specified in 3.2.8, the solenoid shall operate in either a continuous or intermittent duty cycle as specified (see 6.1), at 85 percent of rated voltage and drop out above 10 percent of rated voltage.

3.3.2 Voltage. The solenoids shall operate at one of the following voltages:

(a) 440 volts single phase 60 Hertz.
(b) 115 volts single phase 60 Hertz.
(c) 230 volts direct current (d.c.).
(d) 250 volts d.c. Nominal (210 to 355 volts d.c. submarine service).
(e) other, when specified (see 6.1).

3.3.2.1 Alternating current solenoids shall be designed to operate on type I power of MIL-STD-761, except they shall be limited to 60 Hertz plus or minus 5 percent.

3.3.3 Plunger stroke and pull force shall be as specified (see 6.1), except when the solenoid is part of a valve, the solenoid shall be compatible with the valve.

3.3.4 Dielectric strength. The dielectric withstanding voltages shall be in accordance with NEMA Standard IC 1-2.42.

3.3.5 Insulation resistance. The insulation resistance of coil to ground shall be 10 megohms minimum.

3.3.6 Coil resistance. The resistance on d.c. coils shall be within 10 percent of the design value at 25°C.

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3.4 Technical manuals. Technical manuals shall be in accordance with type I of MIL-M-15071 and MIL-M-81203.

3.5 Identification plates. An identification plate in accordance with MIL-P-15024, stating voltage and frequency, shall be permanently affixed to each solenoid enclosure or frame (when not enclosed).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The inspection system which the supplier is required to maintain, as provided in the inspection clause of the contract or order, shall be in accordance with MIL-I-45208.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

First Article inspection.
Quality conformance inspection.

4.3 First article inspection. When specified (see 6.1), a minimum of three units shall be subjected to the following:

Visual examination	4.5.1
Creepage and clearance test	4.5.2
D.c. resistance	4.5.3
Operating	4.5.4
Temperature rise	4.5.5
Vibration	4.5.6
Insulation resistance	4.5.7
Dielectric tests	4.5.8
Shock	4.5.9
Enclosure (except explosion proof)	4.5.10
Reliability	4.5.11

4.4 Quality conformance inspection. Every solenoids offered for delivery at one time shall be subjected to the following:

D.c. resistance	4.5.3
Operating test	4.5.4
Insulation resistance test	4.5.7
Dielectric tests	4.5.8

4.5 Test procedures.

4.5.1 Visual examination. The solenoids shall be visually examined for conformance with specified materials and for good workmanship.

4.5.2 Creepage and clearance. Creepage and clearance distance shall be demonstrated by actual measurement in accordance with MIL-E-917.

4.5.3 D.c. resistance. The resistance of d.c. solenoid coils shall be measured to determine conformance with 3.3.6.

4.5.4 Operating. Solenoids shall be tested to determine conformance with 3.3.1 and 3.3.2.

4.5.5 Temperature rise. Tests shall be made at specified rated voltages and applicable duty cycle (see 3.3.1 and 3.3.2) to determine conformance with 3.2.7.

4.5.6 Vibration. Solenoids shall be checked at each frequency of the type I vibration test of MIL-STD-167.

4.5.7 Insulation resistance. The insulation resistance of the solenoids shall be measured with an insulation resistance indicating meter with a full scale reading of 100 or 200 megohms, an open circuit voltage of 500 volts and a voltage of at least 450 volts across a resistance of 1 megohm to determine conformance with 3.3.5.

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4.5.8 Dielectric. The dielectric tests shall be conducted in accordance with NEMA Standard IC 1-2.42 and shall be made after completion of all other tests, except the shock test.

4.5.9 Shock. The solenoids shall be subjected to a shock test in accordance with MIL-S-901 to determine conformance with 3.2.3.

4.5.10 Enclosures. The solenoid enclosures shall be tested in accordance with MIL-STD-108 to determine conformance with 3.2.5.

4.5.11 Reliability. During the operating tests (see 4.5.4), the equipment shall be monitored to assure that there are not more than two failures during the 3,000,000 cycles of operation of the three samples for First article inspection. (Three units x 1,000,000 cycles = 3,000,000 cycles.) A failure shall be construed as any operation where the solenoid does not perform its intended function. If more than two failures occur, the operating test shall be stopped, the failure(s) shall be corrected and three different samples shall then be tested with not more than two failures allowed. If there are more than two failures, the Power Distribution Section, Naval Ship Engineering Center shall be consulted prior to any further action. Data on the reliability tests shall be forwarded with the First article inspection report and shall contain the number and modes of failure(s), failure analysis, and what corrective action was taken.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of reference documents listed in section 2, see 6.2.)

5.1 Domestic shipment and early equipment installation.

5.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules, or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment requirements. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

5.3 Use of polystyrene (loose-fill) material.

5.3.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.1), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellosic material bound fiber, fiberboard or transparent flexible cellular material."

5.3.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Samples for First article inspection (see 3.1).

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(b) Type of enclosure, when required (see 3.2.5).
(c) Electrical operations, if other than specified (see 3.3.1).
(d) Duty cycle (see 3.3.1).
(e) Voltage, when other than specified (see 3.3.2).
(f) Plunger stroke and pull force (see 3.3.3).
(g) Levels of preservation, packaging, packing, and marking (see 5.1 and 5.2).
(h) Approval of polystyrene material (see 5.3).

6.2 Sub-contracted material and parts. The preparation for delivery requirements of reference documents cited in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.3 Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy-SH
(Project 6110-N049)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0101-807-2000

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